# Integrated Software Project

## Assignment 4

### Functional Prototype and Presentation

| Assessment information | Details |
|---|---|
| Module code | ISJ107V |
| Assessment weighting | 20% of the final module mark |
| Total marks | 50 marks |
| Presentation period | Presentations commence in early October 2026 |
| Assessment type | Individual |
| Submission | Presentation starts from 3rd October 2026 |

### Instructions

- Use the same approved project and emerging technology from Assignment 1, 2 & 3 unless the lecturer approved a change.
- Presentations commence in 3rd October 2026. The lecturer will communicate the detailed timetable
- Each student must attend the allocated session; demonstrate the prototype live and answer questions about the design and implementation.
- Use a recognised UML tool. Diagrams must be readable, labelled, internally consistent and aligned with the prototype.
- Submit one searchable PDF named ISJ107V_Assignment4_StudentNumber.pdf, presentation slides through Brightspace (MyTUTor)
- Keep the report between 8 and 12 pages excluding the cover, references and appendices. Include captions, references, and declarations of original work and permitted AI-tool use.

## Question 1 — Total Marks [9]

Define what the prototype demonstrates and establish its readiness for assessment.

1.1 Provide the project title, problem addressed, intended users, prototype purpose and selected emerging technology. (2)

1.2 Identify at least 5 functional requirements demonstrated by the prototype and distinguish implemented, partially implemented and deferred features. (5)

1.3 Provide clear setup and access instructions, including required software, hardware, test credentials or sample data and the repository location. (2)

## Question 2 — Total Marks [16]

Develop a working prototype that demonstrates the central value of the proposed system.

2.1 Implement one complete end-to-end user workflow with appropriate navigation, input validation, meaningful output and error feedback. (6)

2.2 Integrate the approved emerging-technology component into the workflow. Demonstrate that it receives valid input and produces an observable result used by the system. (5)

2.3 Implement persistent or realistic sample data through a database, dataset, API, sensor, ledger or suitable simulated service. (3)

2.4 Apply basic security and reliability controls, including access control where relevant, safe error handling and protection of credentials or personal data. (2)

Recommended prototype evidence format:

| Requirement | Status | Prototype feature | Test case | Evidence |
|---|---|---|---|---|
| FR-01 | Completed | dashboard | TC-01 | Screenshot / live demo |
| ETR-01 | Completed | Emerging-tech service | TC-05 | Metric / output |

## Question 3 Total Marks [10]

Use UML to explain the structure and behaviour of the demonstrated prototype.

3.1 Provide an updated use-case diagram identifying the actors and prototype functions included in the demonstration. (2)

3.2 Provide an activity diagram for the demonstrated workflow, including decisions, alternative paths and the emerging technology step. (2)

3.3 Provide an updated class diagram showing the main prototype classes, attributes, operations, associations and multiplicities. (2)

3.4 Provide a sequence diagram showing the interaction between the user interface, application logic, data service and emerging-technology component. (2)

3.5 Provide a component or deployment diagram showing the prototype modules and execution environment. Explain any important difference between the UML design and the implemented prototype. (2)

## Question 4 Total Marks [8]

Provide evidence that the prototype works and identifies its present limitations.

4.1 Test at least five representative cases covering the main workflow, invalid input, data or API interaction and the emerging-technology component. Report expected result, actual result and status. (4)

4.2 Report at least two quantitative measures appropriate to the prototype, such as response time, accuracy, task completion, reliability or usability. (2)

4.3 Identify two important prototype limitations or risks and propose realistic corrective actions for the final system. (2)

## Question 5 Prototype presentation Total Marks [10]

Present the prototype professionally and demonstrate individual understanding of the project.

5.1 Deliver a structured presentation of 5 minutes covering the problem, proposed solution, architecture, emerging technology, prototype features, results and limitations. (3)

5.2 Conduct a reliable live demonstration (10 minutes) of the main workflow and emerging-technology integration using prepared sample data and a fallback plan. (4)

5.3 Use clear, readable presentation slides and communicate professionally within the allocated time. (1)

5.4 Answer technical questions accurately and explain individual design, implementation and testing decisions. (2)

## Assessment rubric

| Criterion | Excellent | Competent | Developing | Insufficient |
|---|---|---|---|---|
| Question 1<br>6 marks | 6<br>Scope, feature status, setup and access details are complete, accurate and demonstration ready. | 4-5<br>Prototype readiness is clear, with minor omissions in status or setup information. | 2-3<br>Basic scope is stated, but readiness, access or feature status is unclear. | 0-1<br>Prototype scope and access information are substantially missing. |
| Question 2<br>16 marks | 14-16<br>A stable end-to-end prototype integrates emerging technology, realistic data and appropriate controls. | 10-13<br>Core workflow and integration work, with minor functional, data or reliability gaps. | 6-9<br>Prototype is partly functional but workflow or emerging-technology integration is limited. | 0-5<br>Prototype is largely non-functional, disconnected or lacks the required integration. |
| Question 3<br>10 marks | 9-10<br>All UML models are correct, readable, consistent and accurately aligned with the prototype. | 6-8<br>Models explain the prototype well, with minor notation or consistency errors. | 3-5<br>Most models are present but simplified, inconsistent or contain several errors. | 0-2<br>Models are missing, unreadable or unrelated to the prototype. |
| Question 4<br>8 marks | 7-8<br>Testing is credible and evidence-based; metrics, limitations and corrective actions are appropriate. | 5-6<br>Testing covers most required areas, with limited evidence or evaluation in one area. | 3-4<br>Some testing is shown, but cases, metrics or corrective actions are incomplete. | 0-2<br>Little credible testing or evaluation evidence is supplied. |

| Criterion | Excellent | Competent | Developing | Insufficient |
|---|---|---|---|---|
| Question 5<br>10 marks | 9-10<br>Presentation is focused and professional; live demo is convincing and technical answers show strong individual understanding. | 6-8<br>Clear presentation and working demo, with minor timing, communication or response weaknesses. | 3-5<br>Presentation or demo is incomplete, unclear or weakly supported during questioning. | 0-2<br>Presentation is unprepared, prototype cannot be demonstrated, or understanding is not evident. |

**Marker summary Q**1 ___ /6 Q2 ____ /16 Q3 ____ /10 Q4 ____ /8 Q5 ____ /10 **TOTAL ____ /50**

**Marker comments** _____